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Content

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Exploiting insecure deserialization vulnerabilities

In this section, we'll teach you how to exploit some common scenarios using examples from PHP, Ruby, and Java deserialization. We hope to demonstrate how exploiting insecure deserialization is actually much easier than many people believe. This is even the case during blackbox testing if you are able to use pre-built gadget chains.

We'll also guide you through the process of creating your own high-severity deserialization-based attacks. Although these usually require source-code access, they can also be easier to learn than you might think once you understand the basic concepts. In particular, we'll cover the following topics:

- [How to identify insecure deserialization LABS](#)
- [Modifying serialized objects that are expected by the website LABS](#)
- [Passing malicious data into dangerous website functionality LABS](#)
- [Injecting arbitrary object types LABS](#)
- [Chaining method invocations to control the flow of data into dangerous sink gadgets LABS](#)
- [Manually creating your own advanced exploits LABS](#)
- [PHAR deserialization LABS](#)

Note

Although many of the labs and examples are based on PHP, most of the exploitation techniques are equally valid for other languages as well.

How to identify insecure deserialization

Identifying insecure deserialization is relatively simple regardless of whether you are whitebox or blackbox testing.

During auditing, you should look at all data being passed into the website and try to identify anything that looks like serialized data. Serialized data can be identified relatively easily if you know the format that different languages use. In this section, we'll show examples from both PHP and Java serialization. Once you identify serialized data, you can test whether you are able to control it.

Tip

For users of Burp Suite Professional, Burp Scanner will automatically flag any HTTP messages that appear to contain serialized objects.

PHP serialization format

PHP uses a mostly human-readable string format, with letters representing the data type and numbers representing the length of each entry. For example, consider a `User` object with the attributes:

```
$user->name = "carlos"; $user->isLoggedIn = true;
```

When serialized, this object may look something like this:

```
0:4:"User":2:{s:4:"name":s:6:"carlos";s:10:"isLoggedIn":b:1;}
```

This can be interpreted as follows:

- `0:4:"User"` - An object with the 4-character class name `"User"`
- `2` - the object has 2 attributes
- `s:4:"name"` - The key of the first attribute is the 4-character string `"name"`
- `s:6:"carlos"` - The value of the first attribute is the 6-character string `"carlos"`
- `s:10:"isLoggedIn"` - The key of the second attribute is the 10-character string `"isLoggedIn"`
- `b:1` - The value of the second attribute is the boolean value `true`

The native methods for PHP serialization are `serialize()` and `unserialize()`. If you have source code access, you should start by looking for `unserialize()` anywhere in the code and investigating further.

Java serialization format

Some languages, such as Java, use binary serialization formats. This is more difficult to read, but you can still identify serialized data if you know how to recognize a few tell-tale

signs. For example, serialized Java objects always begin with the same bytes, which are encoded as `ac ed` in hexadecimal and `r00` in Base64.

Any class that implements the interface `java.io.Serializable` can be serialized and deserialized. If you have source code access, take note of any code that uses the `readObject()` method, which is used to read and deserialize data from an `InputStream`.

Manipulating serialized objects

Exploiting some deserialization vulnerabilities can be as easy as changing an attribute in a serialized object. As the object state is persisted, you can study the serialized data to identify and edit interesting attribute values. You can then pass the malicious object into the website via its deserialization process. This is the initial step for a basic deserialization exploit.

Broadly speaking, there are two approaches you can take when manipulating serialized objects. You can either edit the object directly in its byte stream form, or you can write a short script in the corresponding language to create and serialize the new object yourself. The latter approach is often easier when working with binary serialization formats.

Modifying object attributes

When tampering with the data, as long as the attacker preserves a valid serialized object, the deserialization process will create a server-side object with the modified attribute values.

As a simple example, consider a website that uses a serialized `User` object to store data about a user's session in a cookie. If an attacker spotted this serialized object in an HTTP request, they might decode it to find the following byte stream:

```
0:4:"User":2:{s:8:"username";s:6:"carlos";s:7:"isAdmin";b:0;}
```

The `isAdmin` attribute is an obvious point of interest. An attacker could simply change the boolean value of the attribute to `1` (true), re-encode the object, and overwrite their current cookie with this modified value. In isolation, this has no effect. However, let's say the website uses this cookie to check whether the current user has access to certain administrative functionality:

```
$user = unserialize($_COOKIE); if ($user->isAdmin === true) { // allow access to admin interface }
```

This vulnerable code would instantiate a `User` object based on the data from the cookie, including the attacker-modified `isAdmin` attribute. At no point is the authenticity of the serialized object checked. This data is then passed into the conditional statement and, in this case, would allow for an easy privilege escalation.

This simple scenario is not common in the wild. However, editing an attribute value in this way demonstrates the first step towards accessing the massive amount of attack-surface exposed by insecure deserialization.

Modifying data types

We've seen how you can modify attribute values in serialized objects, but it's also possible to supply unexpected data types.

PHP-based logic is particularly vulnerable to this kind of manipulation due to the behavior of its loose comparison operator (`==`) when comparing different data types. For example, if you perform a loose comparison between an integer and a string, PHP will attempt to convert the string to an integer, meaning that `5 == "5"` evaluates to `true`.

Unusually, this also works for any alphanumeric string that starts with a number. In this case, PHP will effectively convert the entire string to an integer value based on the initial number. The rest of the string is ignored completely. Therefore, `5 == "5 of something"` is in practice treated as `5 == 5`.

Likewise, on PHP 7.x and earlier the comparison `0 == "Example string"` evaluates to `true`, because PHP treats the entire string as the integer `0`.

Consider a case where this loose comparison operator is used in conjunction with user-controllable data from a deserialized object. This could potentially result in dangerous logic flaws.

```
$login = unserialize($_COOKIE) if ($login['password'] == $password) { // log in successfully }
```

Let's say an attacker modified the password attribute so that it contained the integer `0` instead of the expected string. As long as the stored password does not start with a number, the condition would always return `true`, enabling an authentication bypass. Note that this is only possible because deserialization preserves the data type. If the code fetched the password from the request directly, the `0` would be converted to a string and the condition would evaluate to `false`.

Note

In PHP 8 and later, the `0 == "Example string"` comparison evaluates to `false` because strings are no longer implicitly converted to `0` during comparisons. As a result, this exploit is not possible on these versions of PHP.

The behavior when comparing an alphanumeric string that starts with a number remains the same in PHP 8. As such, `5 == "5 of something"` is still treated as `5 == 5`.

Be aware that when modifying data types in any serialized object format, it is important to remember to update any type labels and length indicators in the serialized data too. Otherwise, the serialized object will be corrupted and will not be deserialized.

When working directly with binary formats, we recommend using the Hackvector extension, available from the BApp store. With Hackvector, you can modify the serialized data as a string, and it will automatically update the binary data, adjusting the offsets accordingly. This can save you a lot of manual effort.

Using application functionality

As well as simply checking attribute values, a website's functionality might also perform dangerous operations on data from a deserialized object. In this case, you can use insecure deserialization to pass in unexpected data and leverage the related functionality to do damage.

For example, as part of a website's "Delete user" functionality, the user's profile picture is deleted by accessing the file path in the `$user->image_location` attribute. If this `$user` was created from a serialized object, an attacker could exploit this by passing in a modified object with the `image_location` set to an arbitrary file path. Deleting their own user account would then delete this arbitrary file as well.

This example relies on the attacker manually invoking the dangerous method via user-accessible functionality. However, insecure deserialization becomes much more interesting when you create exploits that pass data into dangerous methods automatically. This is enabled by the use of "magic methods".

Magic methods

Magic methods are a special subset of methods that you do not have to explicitly invoke. Instead, they are invoked automatically whenever a particular event or scenario occurs. Magic methods are a common feature of object-oriented programming in various languages. They are sometimes indicated by prefixing or surrounding the method name with double-underscores.

Developers can add magic methods to a class in order to predetermine what code should be executed when the corresponding event or scenario occurs. Exactly when and why a magic method is invoked differs from method to method. One of the most common examples in PHP is `__construct()`, which is invoked whenever an object of the class is instantiated, similar to Python's `__init__`. Typically, constructor magic methods like this contain code to initialize the attributes of the instance. However, magic methods can be customized by developers to execute any code they want.

Magic methods are widely used and do not represent a vulnerability on their own. But they can become dangerous when the code that they execute handles attacker-controllable data, for example, from a deserialized object. This can be exploited by an attacker to automatically invoke methods on the deserialized data when the corresponding conditions are met.

Most importantly in this context, some languages have magic methods that are invoked automatically **during** the deserialization process. For example, PHP's `unserialize()` method looks for and invokes an object's `__wakeup()` magic method.

In Java deserialization, the same applies to the `ObjectInputStream.readObject()` method, which is used to read data from the initial byte stream and essentially acts like a constructor for "re-initializing" a serialized object. However, `Serializable` classes can also declare their own `readObject()` method as follows:

```
private void readObject(ObjectInputStream in) throws IOException,  
ClassNotFoundException { // implementation }
```

A `readObject()` method declared in exactly this way acts as a magic method that is invoked during deserialization. This allows the class to control the deserialization of its own fields more closely.

You should pay close attention to any classes that contain these types of magic methods. They allow you to pass data from a serialized object into the website's code before the object is fully deserialized. This is the starting point for creating more advanced exploits.

Injecting arbitrary objects

As we've seen, it is occasionally possible to exploit insecure deserialization by simply editing the object supplied by the website. However, injecting arbitrary object types can open up many more possibilities.

In object-oriented programming, the methods available to an object are determined by its class. Therefore, if an attacker can manipulate which class of object is being passed in as serialized data, they can influence what code is executed after, and even during, deserialization.

Deserialization methods do not typically check what they are deserializing. This means that you can pass in objects of any serializable class that is available to the website, and the object will be deserialized. This effectively allows an attacker to create instances of arbitrary classes. The fact that this object is not of the expected class does not matter. The unexpected object type might cause an exception in the application logic, but the malicious object will already be instantiated by then.

If an attacker has access to the source code, they can study all of the available classes in detail. To construct a simple exploit, they would look for classes containing deserialization magic methods, then check whether any of them perform dangerous operations on controllable data. The attacker can then pass in a serialized object of this class to use its magic method for an exploit.

Classes containing these deserialization magic methods can also be used to initiate more complex attacks involving a long series of method invocations, known as a "gadget chain".

Gadget chains

A "gadget" is a snippet of code that exists in the application that can help an attacker to achieve a particular goal. An individual gadget may not directly do anything harmful with user input. However, the attacker's goal might simply be to invoke a method that will pass their input into another gadget. By chaining multiple gadgets together in this way, an attacker can potentially pass their input into a dangerous "sink gadget", where it can cause maximum damage.

It is important to understand that, unlike some other types of exploit, a gadget chain is not a payload of chained methods constructed by the attacker. All of the code already exists on the website. The only thing the attacker controls is the data that is passed into the

gadget chain. This is typically done using a magic method that is invoked during deserialization, sometimes known as a "kick-off gadget".

In the wild, many insecure deserialization vulnerabilities will only be exploitable through the use of gadget chains. This can sometimes be a simple one or two-step chain, but constructing high-severity attacks will likely require a more elaborate sequence of object instantiations and method invocations. Therefore, being able to construct gadget chains is one of the key aspects of successfully exploiting insecure deserialization.

Working with pre-built gadget chains

Manually identifying gadget chains can be a fairly arduous process, and is almost impossible without source code access. Fortunately, there are a few options for working with pre-built gadget chains that you can try first.

There are several tools available that provide a range of pre-discovered chains that have been successfully exploited on other websites. Even if you don't have access to the source code, you can use these tools to both identify and exploit insecure deserialization vulnerabilities with relatively little effort. This approach is made possible due to the widespread use of libraries that contain exploitable gadget chains. For example, if a gadget chain in Java's Apache Commons Collections library can be exploited on one website, any other website that implements this library may also be exploitable using the same chain.

ysoserial

One such tool for Java deserialization is "ysoserial". This lets you choose one of the provided gadget chains for a library that you think the target application is using, then pass in a command that you want to execute. It then creates an appropriate serialized object based on the selected chain. This still involves a certain amount of trial and error, but it is considerably less labor-intensive than constructing your own gadget chains manually.

Note

In Java versions 16 and above, you need to set a series of command-line arguments for Java to run ysoserial. For example:

```
java -jar ysoserial-all.jar \ --add-opens=java.xml/com.sun.org.apache.xalan.internal.xsltc.trax=ALL-UNNAMED \ --add-opens=java.xml/com.sun.org.apache.xalan.internal.xsltc.runtime=ALL-UNNAMED \ --add-opens=java.base/java.net=ALL-UNNAMED \ --add-opens=java.base/java.util=ALL-UNNAMED \ [payload] '[command]'
```

Not all of the gadget chains in ysoserial enable you to run arbitrary code. Instead, they may be useful for other purposes. For example, you can use the following ones to help you quickly detect insecure deserialization on virtually any server:

- The `URLDNS` chain triggers a DNS lookup for a supplied URL. Most importantly, it does not rely on the target application using a specific vulnerable library and works in any

known Java version. This makes it the most universal gadget chain for detection purposes. If you spot a serialized object in the traffic, you can try using this gadget chain to generate an object that triggers a DNS interaction with the Burp Collaborator server. If it does, you can be sure that deserialization occurred on your target.

- `JRMPCClient` is another universal chain that you can use for initial detection. It causes the server to try establishing a TCP connection to the supplied IP address. Note that you need to provide a raw IP address rather than a hostname. This chain may be useful in environments where all outbound traffic is firewalled, including DNS lookups. You can try generating payloads with two different IP addresses: a local one and a firewalled, external one. If the application responds immediately for a payload with a local address, but hangs for a payload with an external address, causing a delay in the response, this indicates that the gadget chain worked because the server tried to connect to the firewalled address. In this case, the subtle time difference in responses can help you to detect whether deserialization occurs on the server, even in blind cases.

PHP Generic Gadget Chains

Most languages that frequently suffer from insecure deserialization vulnerabilities have equivalent proof-of-concept tools. For example, for PHP-based sites you can use "PHP Generic Gadget Chains" (PHPGGC).

Note

It is important to note that the vulnerability is the deserialization of user-controllable data, not the mere presence of a gadget chain in the website's code or any of its libraries. The gadget chain is just a means of manipulating the flow of the harmful data once it has been injected. This also applies to various memory corruption vulnerabilities that rely on deserialization of untrusted data. In other words, a website may still be vulnerable even if it did somehow manage to plug every possible gadget chain.

Working with documented gadget chains

There may not always be a dedicated tool available for exploiting known gadget chains in the framework used by the target application. In this case, it's always worth looking online to see if there are any documented exploits that you can adapt manually. Tweaking the code may require some basic understanding of the language and framework, and you might sometimes need to serialize the object yourself, but this approach is still considerably less effort than building an exploit from scratch.

Even if you can't find a gadget chain that's ready to use, you may still gain valuable knowledge that helps you create your own custom exploit.

Creating your own exploit

When off-the-shelf gadget chains and documented exploits are unsuccessful, you will need to create your own exploit.

To successfully build your own gadget chain, you will almost certainly need source code access. The first step is to study this source code to identify a class that contains a magic method that is invoked during deserialization. Assess the code that this magic method executes to see if it directly does anything dangerous with user-controllable attributes. This is always worth checking just in case.

If the magic method is not exploitable on its own, it can serve as your "kick-off gadget" for a gadget chain. Study any methods that the kick-off gadget invokes. Do any of these do something dangerous with data that you control? If not, take a closer look at each of the methods that they subsequently invoke, and so on.

Repeat this process, keeping track of which values you have access to, until you either reach a dead end or identify a dangerous sink gadget into which your controllable data is passed.

Once you've worked out how to successfully construct a gadget chain within the application code, the next step is to create a serialized object containing your payload. This is simply a case of studying the class declaration in the source code and creating a valid serialized object with the appropriate values required for your exploit. As we have seen in previous labs, this is relatively simple when working with string-based serialization formats.

Working with binary formats, such as when constructing a Java deserialization exploit, can be particularly cumbersome. When making minor changes to an existing object, you might be comfortable working directly with the bytes. However, when making more significant changes, such as passing in a completely new object, this quickly becomes impractical. It is often much simpler to write your own code in the target language in order to generate and serialize the data yourself.

When creating your own gadget chain, look out for opportunities to use this extra attack surface to trigger secondary vulnerabilities.

By carefully studying the source code, you can discover longer gadget chains that potentially allow you to construct high-severity attacks, often including remote code execution.

PHAR deserialization

So far, we've looked primarily at exploiting deserialization vulnerabilities where the website explicitly deserializes user input. However, in PHP it is sometimes possible to exploit deserialization even if there is no obvious use of the `unserialize()` method.

PHP provides several URL-style wrappers that you can use for handling different protocols when accessing file paths. One of these is the `phar://` wrapper, which provides a stream interface for accessing PHP Archive (`.phar`) files.

The PHP documentation reveals that PHAR manifest files contain serialized metadata. Crucially, if you perform any filesystem operations on a `phar://` stream, this metadata is implicitly deserialized. This means that a `phar://` stream can potentially be a vector for

exploiting insecure deserialization, provided that you can pass this stream into a `filesystem` method.

In the case of obviously dangerous filesystem methods, such as `include()` or `fopen()`, websites are likely to have implemented counter-measures to reduce the potential for them to be used maliciously. However, methods such as `file_exists()`, which are not so overtly dangerous, may not be as well protected.

This technique also requires you to upload the `PHAR` to the server somehow. One approach is to use an image upload functionality, for example. If you are able to create a polyglot file, with a `PHAR` masquerading as a simple `JPG`, you can sometimes bypass the website's validation checks. If you can then force the website to load this polyglot "`JPG`" from a `phar://` stream, any harmful data you inject via the `PHAR` metadata will be deserialized. As the file extension is not checked when PHP reads a stream, it does not matter that the file uses an image extension.

As long as the class of the object is supported by the website, both the `__wakeup()` and `__destruct()` magic methods can be invoked in this way, allowing you to potentially kick off a gadget chain using this technique.

This inventive technique was featured in our Top 10 web hacking techniques of 2018.

Read more

- [Top 10 web hacking techniques of 2018](#)

Exploiting deserialization using memory corruption

Even without the use of gadget chains, it is still possible to exploit insecure deserialization. If all else fails, there are often publicly documented memory corruption vulnerabilities that can be exploited via insecure deserialization. These typically lead to remote code execution.

Deserialization methods, such as PHP's `unserialize()` are rarely hardened against these kinds of attacks, and expose a huge amount of attack surface. This is not always considered a vulnerability in its own right because these methods are not intended to handle user-controllable input in the first place.